SAAD BABEZZINE

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SUMMARY

Electrical engineering junior at UMass Boston seeking a summer 2027 internship in power, energy, or MEP electrical design. I studied transformers for my Circuit Analysis II final project, led a team that built a solar power system, and I am now designing and testing a regulated power supply. I am training in AutoCAD Electrical and want to learn how electrical systems for buildings and the grid are planned, designed, and built.

EDUCATION

University of Massachusetts Boston, Boston, MA — Expected May 2028

- Bachelor of Science in Electrical Engineering, junior standing
- Current courses include Electronics I with Lab, Signals and Systems, Engineering Mathematics, Introduction to Electrical and Computer Engineering

Bunker Hill Community College, Boston, MA — May 2026

- Associate degree, Engineering Transfer Program, GPA 3.94
- Completed courses include Circuit Analysis I and II, Digital Logic Design, Differential Equations, Physics I and II

ENGINEERING PROJECTS

Transformer Project, Circuit Analysis II Final Project, Bunker Hill Community College — Spring 2026

- Researched how transformers step voltage up and down using the turns ratio
- Analyzed voltage, current, and power on both sides of a transformer
- Connected what I learned in class to how power actually gets delivered to homes and businesses

Solar Power System Project, Bunker Hill Community College

- Led a team of three to design and build a solar power system, dividing the work and keeping the project on schedule
- Presented the finished system and our results to the class
- Learned that leading means checking in early, matching tasks to people's strengths, and keeping everyone on one plan

Op Amp Amplifier Design and Testing, Electronics I, UMass Boston — Fall 2026

- Built and tested inverting and noninverting LM741 amplifiers, confirming designed gains of 10 and 11 on the oscilloscope
- Measured 3 dB bandwidth at gains of 10, 30, and 100 (86 kHz, 29 kHz, 9.5 kHz) and showed the gain bandwidth product stayed near the 1 MHz datasheet value
- Found the output clipping point and fixed a source loading error by matching the function generator load setting
- Learned that real parts never behave exactly like the ideal model, and how to use the datasheet to explain the difference

Power Quality Harmonic Analyzer, Signals and Systems, UMass Boston — Fall 2026, in progress

- Breaking the distorted current drawn by everyday electronics into harmonics with Fourier analysis in MATLAB and calculating total harmonic distortion
- Goal is to see how much distortion common chargers and LED bulbs put back onto the power line

Regulated DC Power Supply, Electronics I, UMass Boston — Fall 2026, in progress

- Designing a supply with a diode bridge rectifier, filter capacitor, and regulator stage, and testing ripple under different loads
- Goal is to understand every stage between an AC input and a steady DC output

Solar Tracker, Intro to Electrical and Computer Engineering, UMass Boston — Fall 2026, in progress

- Building a light sensor and servo system that turns a small solar panel toward the light, with a mount designed in CAD and 3D printed
- Goal is to measure how much more light a tracking panel collects than a fixed one

EXPERIENCE

Package Car Driver (Casual), UPS, Watertown, MA — Jun 2026 to Sep 2026

- Completed UPS Integrad safety training, passed probation, and ran routes of up to 170 stops a day on strict time commitments

Lab Assistant, Bunker Hill Community College, Boston, MA — Feb 2026 to May 2026

- Helped students set up lab equipment and work through circuit experiments, and kept the lab organized and safe
- Explaining circuits to others made me understand them better and taught me patience when someone is stuck

Delivery Driver, Amazon Delivery Service Partner — May 2024 to May 2025

- Delivered 350 to 450 packages across 150 to 190 stops per route, and trained a new driver on the route and station procedures

TECHNICAL SKILLS

- Circuit design and analysis, analog and digital circuits, op amp circuits, AC and DC circuits, phasor analysis, semiconductor fundamentals
- Power systems fundamentals, transformers, electrical design, and technical writing for lab reports
- Laboratory testing with oscilloscope, function generator, digital multimeter, and DC power supply, breadboard prototyping
- CAD, rapid prototyping with 3D printing, AutoCAD Electrical (training in progress), MATLAB for Fourier analysis
- C++, HTML and CSS, Microsoft Visual Studio, Excel, Word, PowerPoint, teamwork, and safety procedures